PRODUCT DOCUMENT · SIX TABS

Supplier Follow-up Agent: product document

The thinking behind the prototype: the problem and decision, the spec, AI and evaluation, metrics and launch, and the risks with the build and decision logs.

Supplier Follow-up Agent

Oct 11, 2026 · @Rohit Kumar

Proposal: add a follow-up agent to GST reconciliation that gets suppliers to fix their filings before the credit deadline. It is a working prototype on fictional data, made by Rohit Kumar using Claude; no users have tested it yet.

At a glance

Problem	Reconciliation finds invoices that are missing or wrong in GSTR-2B, but only the supplier can fix them. Chasing runs on email, WhatsApp and spreadsheets, so promises slip and last year's credit expires on 30 November.
User	The GST lead at a mid-size buyer with several GSTINs. Persona: Abhishek Attri, GST lead at Kestrel Auto Components (fictional).
Product	An inbox ranked by what would be lost first, and one page per supplier with what is open, what they promised and the proof. Rules choose every step, AI reads replies and writes drafts, code checks every AI answer, and a person approves what goes out.
Why AI is essential	Supplier replies are free text, often in Hindi, Hinglish or a regional language; the test covered five languages. Without AI, a person reads every reply.
North Star	Credit recovered, counted only when the invoice appears in GSTR-2B.
Demo result (simulated)	Taking the recommended steps on 15 Oct recovers ₹22.4 L by 14 Nov and loses nothing. Doing nothing recovers ₹6.42 L and loses ₹6.26 L of FY 2025-26 credit for good.
AI test	40 replies in five languages, run blind: the quick model got 40 of 40 positions and dates right and flagged 19 of 20 replies that need a person. The set was written with Claude, so this shows consistency, not accuracy on real replies.

Not proven	Real users, real replies, and the baseline loss rate, which drives most of the value.
First step	Two weeks: interview five GST leads and have tax experts label 300 real, anonymised replies.

Working-backwards note (a proposal, not an announcement)

Written as the launch note this feature would have to earn. It describes the intended product; nothing here is a commitment, and there are no customer quotes because no customer has used it yet.

GST teams get back the credit stuck with suppliers, before the deadline.

A follow-up agent inside reconciliation reads supplier replies in English and Indian languages, tracks every promise, and counts credit as recovered only when it appears in GSTR-2B.

The problem. Reconciliation tells a GST team which invoices are missing or wrong in GSTR-2B. The fix sits with the supplier, who has to report or correct the invoice in their GSTR-1. Teams chase by email and WhatsApp, track promises in spreadsheets, and lose some credit for good when last year's invoices miss the 30 November deadline.

How it works.

- Every flagged supplier lands in one inbox, ranked by what would be lost first.
- The agent reads each reply, quotes the words it relied on, and records the promised date. Anything unclear, disputed, conditional or angry goes to a person.
- Rules the team sets decide every step: reminders before a promised date, follow-ups after a leave, a firmer message when a promise breaks. Messages are drafted in the supplier's language and approved before they go.
- IMS shows what a supplier has uploaded within days; the next GSTR-2B confirms it. Only then does the credit count.

What it will not do. It never sends a message without approval in the first release. It never suggests holding a payment to a micro or small enterprise, or on an invoice older than 150 days. It never counts a supplier's word as recovered credit. It files nothing.

Getting started. It works on the reconciliation a customer already runs. The team connects a follow-up mailbox and, optionally, WhatsApp Business, and reviews the default rules.

FAQ

Will it message my suppliers without my approval? Not in the first release. Every message is approved by a person. Routine reminders can later run on their own, per company, once shadow mode shows the agent agrees with people at least 95% of the time.

What if the AI misreads a reply? Every reading quotes the words it relied on, and code rejects a quote that is not in the message. Disputes, conditions, requests for documents, unclear replies and

angry tones go to a person. Each correction becomes a new test case.

Does it hold supplier payments? Off by default. Switched on, it can only suggest holding the GST part of a payment, never for a micro or small enterprise (paid within 45 days under the MSMED Act) and never on an invoice older than 150 days, because after 180 days unpaid the buyer's own credit is blocked (Rule 37).

Which languages and channels? Email and WhatsApp. English, Hindi, Hinglish, Gujarati and Marathi were tested. On WhatsApp, messages outside the 24-hour reply window go out as pre-approved templates.

How do I know it worked? Only GSTR-2B counts. The dashboard shows credit recovered, open and lost, and every supplier has a case file with the full history.

What is it worth? An illustrative model puts it at about ₹16 L a year for a typical customer against a price of about ₹2 L, with AI costing under ₹1 per supplier per cycle. Every input is an assumption until shadow mode measures the baseline. See Metrics and launch.

Why not just send more reminder emails? Reconciliation can already email suppliers and keep a mail log. What is missing is reading the replies, keeping track of promises and acting on what each supplier said.

Why now? IMS shows a buyer what suppliers have uploaded within days, so the agent can confirm a fix before GSTR-2B arrives. And every November the FY deadline turns slow follow-up into permanent loss.

What has to be true for this to work? Suppliers reply in writing; customers let the agent read a follow-up mailbox; and enough credit is lost or delayed today to matter. The first two weeks test all three.

Links

Start with Assignment_Submission_Rohit_Kumar.pdf: it holds the cover, the write-up and the six slides. The other files sit beside it in the same folder.

What	Where
Submission page	Assignment_Submission_Rohit_Kumar.pdf
Prototype	https://claude.ai/artifact/UzQyvCzPyiLm4JkyeYcA9i (sign-in filled in)
Slides	Slides_Rohit_Kumar.pdf
Build prompt	Build_Prompt_Rohit_Kumar.pdf
Impact model and code	Impact_Model_Rohit_Kumar.xlsx and Source_Code_Rohit_Kumar.zip

Problem and decision

The problem worth solving is not finding mismatches, which reconciliation already does, but getting each supplier to fix them in time.

From request to outcome

Step	Statement
Request	The brief: an AI-enabled feature for a tax product, taken from idea to a working MVP in hours.
Problem	Buyers lose or delay input tax credit when suppliers do not fix the invoices reconciliation flags. Finding the mismatch is solved; closing it with the supplier is manual.
Outcome	More flagged credit recovered within the cycle, no credit lost to the FY deadline, and less of the GST team's time spent chasing.

Problem statement. GST leads at mid-size buyers with several GSTINs need a way to get suppliers to fix GSTR-1 mismatches before the credit deadline, because chasing by email and WhatsApp loses track of who promised what and some credit expires for good. We will know it is solved when credit recovered in GSTR-2B rises and FY credit lost after 30 November falls to zero.

Job to be done. When the reco job flags invoices against a supplier, I want them fixed in that supplier's next GSTR-1 without chasing each one by hand, so I can claim the credit before it lapses and spend my time on the cases that need judgement.

Who has the problem (hypotheses, not yet interviewed)

Three people touch every mismatch. These are working hypotheses from desk research and GST rules, to be tested in interviews.

Role	Wants	Pain today (assumed)	What the product gives them
GST lead (primary user; persona Abhishek Attri)	Claim every rupee of eligible credit, on time, with an audit trail	Chasing dozens of suppliers across email and WhatsApp; promises tracked in spreadsheets; last year's credit noticed too late	One ranked inbox, AI-read replies with evidence, rule-driven reminders, a case file per supplier
Relationship owner (the buyer who manages the supplier)	Keep the supplier relationship healthy and payments on time	Pulled in late, usually when a supplier is already angry or a payment is in question	An alert when tone turns or a promise breaks, with the history already written up

Role	Wants	Pain today (assumed)	What the product gives them
Supplier's accountant	Fix the filing once and be left alone	Vague, repeated reminders in a language or channel they don't use	One specific message in their language: which invoices, what is wrong, by when

When the first version drafted a proto-persona with AI, it assumed that matching takes longer than chasing. I did not accept that; it became the first interview question (H1 below).

Evidence so far

The evidence is desk research. It shows the gap and the deadlines; it does not yet show how much credit customers lose.

Finding	What it means for the product	Source
Masters India's reconciliation offers eight reconciliation types, including IMS vs purchase register, lets users email suppliers about mismatched entries and keeps a mail log. The docs describe no tracking of supplier replies.	The gap sits after the email: reading replies and acting on them.	Masters India docs: Reconciliation
Practitioners advise reconciling often, reminding vendors automatically until they upload, and holding payments for vendors who stay non-compliant.	Reminders are expected; payment holds need guardrails, not a blanket rule.	ClearTax: avoiding lost credit when suppliers don't file
Credit for an invoice cannot be taken after 30 November following the end of the financial year (or the annual return, if earlier).	FY 2025-26 credit must reach the October GSTR-2B: a hard, yearly deadline.	ClearTax: Section 16(4)
Unpaid for 180 days after the invoice, credit must be reversed with interest and is re-availed on payment.	Never suggest holding payment on old invoices: it blocks the buyer's own credit.	ClearTax: Rule 37
If a supplier never files GSTR-3B for an invoice's period, the buyer must reverse the credit (Rule 37A, since December 2022).	Track the supplier's return filing, not just GSTR-1.	ClearTax: Rule 37A
Under IMS, invoices with no action are deemed accepted into GSTR-2B, generated on the 14th.	The agent can see a fix in IMS days before GSTR-2B confirms it.	Tally: how IMS works

Finding	What it means for the product	Source
Payments to micro and small enterprises are deductible only when paid within the MSMED Act limit (45 days at most), under Section 43B(h).	Never suggest holds for micro and small suppliers.	TaxGuru: Section 43B(h)
On WhatsApp, businesses can send free text only within 24 hours of a customer's message; otherwise approved templates.	WhatsApp reminders are templates with details filled in.	Kapso: the 24-hour window

Still missing: interviews, real supplier replies, and any measure of how much flagged credit is lost or delayed today. Sources opened on 11 Oct 2026.

Options considered

AI widened the search; a scorecard narrowed it. The deciding test: if the AI were removed, would the product still work? If yes, the AI is decoration. Scores came from Claude's scoring prompt on desk research (1 = worst, 5 = best); they are judgement, not data.

Option	Pain × frequency	AI is essential	Data already in the product	Low risk if wrong	Measurable in rupees	Total	Decision
Supplier follow-up agent	5: every cycle, every customer	5: replies are free text in five languages	4: reco results exist; needs a follow-up mailbox	4: messages are approved and reversible	5: credit recovered in GSTR-2B	**23**	Chosen
Chat with your dashboard	2	1: filters already answer it	5	4	1: no rupee outcome	13	Dropped. The questions I expect are about one supplier, so the chat moved to each supplier's page.

Option	Pain × frequency	AI is essential	Data already in the product	Low risk if wrong	Measurable in rupees	Total	Decision
Notice-reply drafting assistant	4: rarer but high stakes	4	2: needs the notice, documents and legal judgement	1: legal exposure	3	14	Not now: needs tax law I could not defend in a prototype

Hypotheses to test, and what would make me stop

#	Hypothesis	How to test (first two weeks)	Stop or rethink if
H1	Chasing suppliers takes GST teams longer than finding the mismatches	5 GST-lead interviews: walk through last month's follow-up	Most time goes on matching, not chasing
H2	Replies are too varied for rules: languages, partial answers, conditions	Tax experts label 300 real, anonymised replies	More than 80% of replies are simple "done" or "will do"
H3	Teams will trust an AI reading that shows its evidence	Show the reading screen in interviews; ask what they would check	Leads say they would re-read every reply anyway
H4	At least 3% of flagged supplier-side credit is lost for good today	Ask for last year's reversals and lapsed credit; confirm in shadow mode	Loss is near zero: the value then rests on time saved alone, which may not justify a paid add-on
H5	Suppliers act faster on a specific message in their own language and channel	A/B test in the assisted beta, randomised by supplier	No difference in time to fix

H4 matters most: in the impact model, the loss rate drives most of the value.

Product spec

The agent runs one loop per supplier, from the reco job's mismatch to the GSTR-2B that confirms the fix: rules decide each step, AI reads and writes, and a person approves anything sent.

Goal, non-goals and scope

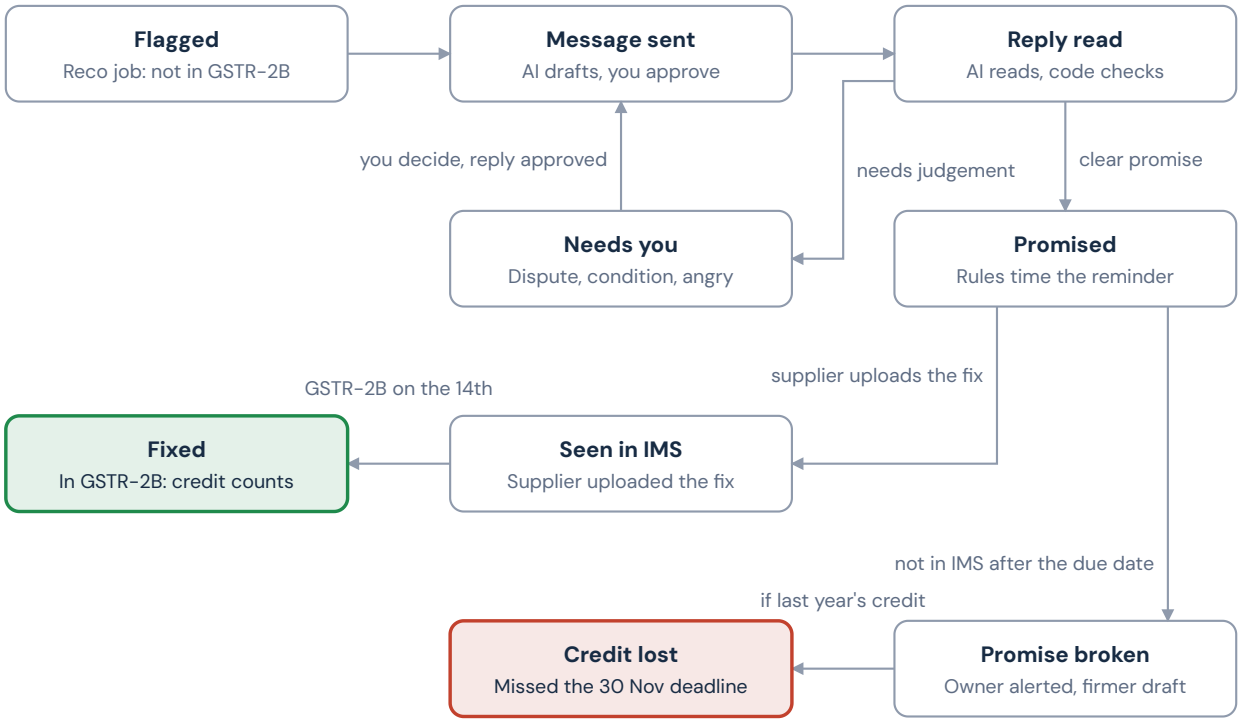
Goal. Get every flagged supplier-side mismatch fixed in the supplier's next GSTR-1, with no credit lost to the FY deadline, while the GST lead handles only the cases that need judgement.

Non-goals for the first release. Filing returns; making or holding payments (the agent may only suggest); reading attachments such as PDFs and scans; a portal for suppliers to log in to; mismatches caused only by our own books (handled, but no messaging).

Priority	Scope
Must	Inbox ranked by what would be lost first · supplier page with history · AI reading of replies with evidence and code checks · AI drafts in the supplier's language, approved before sending · rules with defaults and two locked safety rules · IMS and GSTR-2B checks · append-only history · undo
Should	AI summary per supplier with citations · questions about one supplier · case file PDF · bulk approval of routine messages · dashboard with AI quality · company settings, including AI off
Could	WhatsApp Business templates · payment-hold suggestions (off by default) · supplier filing status from the GST portal · rule templates added by the customer
Won't (now)	Sending without approval · reading attachments · a supplier portal · automatic payment holds

The loop

Credit counts only when GSTR-2B confirms the fix



The follow-up loop · 9 states

A reply that needs judgement goes to a person; a clear promise is timed by the rules. When a promise breaks, the owner is alerted and a firmer message goes back for approval. Only GSTR-2B moves an invoice to Fixed, and last year's credit that misses the October GSTR-2B is lost.

User stories and acceptance criteria

Ten stories make up the first release, and each one has an end-to-end test that runs against the prototype.

Story	Acceptance criteria	Test
1. Ranked inbox. As the GST lead, I want every flagged supplier in one list, ranked by what would be lost first, so I start with what costs most.	Given the reco job has flagged 16 suppliers, when I open the inbox, then the 11 that need a decision come first. Last year's credit leads each group, and every row shows credit stuck, status and next step.	J1, J2
2. Reading with evidence. As the GST lead, I want each reply read for me with the words it relied on, so I can check it in seconds.	Given a supplier's reply, when the AI reads it, then I see its position, invoices, promised date and quoted words. Code drops invoice numbers we don't have and impossible dates; a quote not in the message sends it to a person.	J5, J10

Story	Acceptance criteria	Test
3. Judgement goes to a person. As the GST lead, I want risky replies sent to me, so the agent never answers them on its own.	Given a dispute, a condition, a document request, a wrong contact, an unclear or hostile reply, or low confidence, when it is read, then the supplier moves to Needs you. Nothing goes out automatically, and rule R3 cannot be switched off.	J5, J6
4. Approve, edit, undo. As the GST lead, I want to edit and approve every message and undo a mistake, so nothing goes out unseen.	Given a draft, when I approve it, then I see the checks it passed, and the history records who approved it and whether it was edited. Undo restores the earlier state; the undone send stays in the history.	J3, J4
5. Only GSTR-2B counts. As the GST lead, I want a supplier's "already filed" kept open until the return shows it, so I never claim credit on someone's word.	Given a reply that says an invoice is already filed, when it is read, then the invoice stays open (rule R7, locked). It shows as Seen in IMS once the supplier uploads it, and Fixed only when GSTR-2B arrives on the 14th.	J12
6. Broken promise. As the GST lead, I want a missed promise caught the next day, so it does not slide another month.	Given a supplier promised the fix in the GSTR-1 due on 11 Nov, when the invoice is not in IMS on 12 Nov, then it is marked Promise broken, the owner is alerted and a firmer message waits for my approval.	J12
7. Safe payment holds. As the GST lead, I want a payment hold suggested only where it is safe, so the agent never risks our own credit or MSME law.	Given hold suggestions are on and a supplier will fix only after payment, when the next step is set, then it may suggest holding only the GST part. Never for a micro or small enterprise, never on an invoice over 150 days old; otherwise the page says why.	J4, J9
8. Rule changes with a preview. As the GST lead, I want to see what a rule change will do before I apply it, so suppliers are not surprised.	Given I change a rule's value or add a rule, when I preview it, then each affected supplier's next step shows before and after. Applying it logs the change, and deleting a rule takes a second click.	J8
9. Case file. As the GST lead, I want one supplier's case as a PDF, so I can brief the owner or keep proof for an audit.	Given a supplier page, when I download the case file, then a PDF holds the summary, every invoice, the numbered history and every message, with AI-written text marked.	J11
10. Work without AI. As the GST lead, I want to switch AI off, so the team can still work if we stop trusting it.	Given AI is off in Company settings, when anyone opens a supplier, then the summary is facts only, the reading is hidden with a manual mode, drafts come from templates and questions are off.	J9

J1 to J16 are the automated journeys in the code repository (test/journeys.js); all 73 checks pass. They run on fictional data, so they prove the behaviour, not the value.

Rules

Rules, not the AI, choose every next step. Eight are on by default, two more are templates a team can add, and two can never be switched off.

Rule	When	Then	Default (allowed range)
R1 Clear date promised	The supplier gives a clear date more than a week away	Send one reminder n days before that date	2 days (1 to 7)
R2 Out-of-office reply	An automatic reply gives a return date	Follow up n days after they are back	1 day (1 to 5)
R3 Reply needs judgement	A dispute, a condition, a request for documents, an unclear reply, a wrong contact, a partial answer, a hostile tone or low AI confidence	Send it to a person; never reply automatically	Locked on
R4 Angry or frustrated tone	The AI reads the tone as frustrated or hostile	Alert the relationship owner and slow reminders to every n days	14 days (7 to 30)
R5 Promise broken	The fix is not in IMS after the GSTR-1 due date it was promised for	Mark it broken, alert the owner and draft a firmer message for approval	On
R6 Supplier wants payment first	The supplier will fix only after a payment	Suggest holding only the GST part of the next payment; never for micro or small enterprises or invoices older than n days	150 days (30 to 170); only when payment-hold suggestions are on, which they are not by default
R7 Supplier says it's fixed	The supplier says an invoice is already filed or corrected	Keep it open until it appears in GSTR-2B	Locked on
R8 Last year's credit	The invoice is from FY 2025-26	Flag the 30 Nov 2026 deadline and put it first	On
R9 No reply (template)	No reply n days after our message	Send a reminder	Off; 7 days (3 to 21) when added
R10 Large amount (template)	Credit stuck with the supplier is above x	Every message to them needs approval	Off; ₹2,00,000 when added

The defaults follow the filing calendar and the law. A reminder 2 days before a promise lands before GSTR-1 is due on the 11th. R6 stops at 170 days because credit on an invoice unpaid for 180 days must be reversed (Rule 37), so a hold would hurt the buyer.

R3 and R7 are locked because they guard the two costliest mistakes: an automatic answer to a dispute, and credit counted on a supplier's word. Every rule change is previewed on open suppliers first and kept in a rule history.

Company settings sit above the rules: AI on or off, automatic routine reminders, email and WhatsApp, and payment-hold suggestions. The prototype starts with automatic routine reminders on, to show the end state. The first release ships with them off; a company turns them on only after the agent agrees with its people at least 95% of the time.

Edge cases

Most of the work sits in replies that are not a simple "will fix". The demo gives each common case its own supplier; the last three are in the spec only.

Case	How the agent handles it	In the demo
Supplier says it filed, but GSTR-2B doesn't show it	Kept open (R7). The agent searches the GSTR-2B of every buyer GSTIN; if the invoice sits under another unit, a person decides and the draft asks them to amend the recipient GSTIN.	Patel Steel
Supplier wants documents before amending	Goes to a person (R3); the draft attaches the PO and GRN once approved.	Bharat Fasteners
Fix depends on a payment	Goes to a person, and the owner is alerted and copied. A hold is suggested only within R6's limits; otherwise the draft confirms the scheduled payment date as a fact.	Nova Polymers
The mistake is ours: wrong invoice number in our books	Shown as Fix in our books. No message to the supplier, one approved correction to our entry.	Mehta Castings
Duplicate entry in our purchase register	The supplier says it isn't their bill; after review the duplicate is removed from our books, not chased.	Sai Tooling
Out-of-office reply with a return date	R2: follow up the day after they are back, at the address they gave.	Apex Machine Tools
Contact has left	Goes to a person; the contact is updated and the draft goes to the new address.	GreenLeaf
Vague reply ("ok noted")	Goes to a person; the draft asks for a specific date.	Rathi Alloys
Supplier has stopped filing returns	Their GSTR-3B status is shown with a risk flag: if they never file, our credit must be reversed (Rule 37A).	Vega Chemicals
Angry denial	The owner is alerted (R4); the reply answers invoice by invoice with our evidence, for approval.	Jyoti Industrial Gases

Case	How the agent handles it	In the demo
Promise broken	R5 fires the day after the GSTR-1 due date.	Krishna Electricals
Promise lands after the October GSTR-2B	Tracked to the later month. Fine for this year's credit; last year's credit is tagged "will be lost".	GreenLeaf (December)
Reply in Hindi, Gujarati or Marathi	Read as written; drafts go out in the supplier's language and channel.	Krishna (Hindi); test set
Instructions hidden in a reply	Treated as text: quotes must match, unknown invoice numbers are dropped and nothing is executed.	Test set (E21)
WhatsApp outside the 24-hour window	Sent as the buyer's pre-approved template with the details filled in.	Shree Ganesh
Quarterly filer (QRMP)	Promises are timed to the IFF (by the 13th) or the quarterly GSTR-1, using the supplier's filing frequency.	Spec only
Credit note or amendment after a fix	The next reco run reopens the invoice and links it to the same case.	Spec only
GST portal session about to expire	Warned before the 14th, so the GSTR-2B check is not missed.	Spec only

Data, integrations, permissions and audit

The agent runs on data reconciliation already holds, plus a follow-up mailbox and optional WhatsApp. Every action lands in one append-only log, which is also the audit trail.

Data or integration	Source	Used for
Reco results: GSTR-2B vs PR, IMS vs PR	Masters India reconciliation (exists today)	Which invoices are flagged, and why
IMS and GSTR-2B for every buyer GSTIN	The GST portal connection reconciliation already uses	Seen in IMS, Fixed, and invoices filed to the wrong GSTIN
Supplier return filing (GSTR-1, GSTR-3B)	GST portal	Risk flags (Rule 37A) and timing of promises
Supplier master: contact, language, channel, Udyam category	Customer's vendor master or ERP	Who to write to, in what language; the MSME rule
Payment schedule	Customer's ERP, read only	Payment facts in drafts; the 180-day check

Data or integration	Source	Used for
Follow-up mailbox	Customer's Microsoft 365 or Google Workspace, connected by an admin	Sending approved messages and reading replies
WhatsApp Business (optional)	A WhatsApp Business Platform provider	Suppliers who prefer WhatsApp; templates outside 24 hours
Language model	Claude, through the Anthropic API	Reading, drafting, summaries and questions

Permissions. The GST lead approves messages, confirms or corrects readings, changes rules and downloads case files. A relationship owner sees their suppliers, gets alerts and approves escalations to them. An admin controls company settings and users; an auditor gets read-only history and case files.

Audit. Every entry has a date, an actor (a person, the agent, a rule, the GST portal or the reco job) and the invoices it touches. Nothing is edited or deleted: an undo adds an entry and marks the original undone. AI entries carry the model, prompt version, checks passed and the quoted evidence.

Personal data. Supplier contacts' names, numbers and messages are personal data under the DPDP Act 2023. The customer is the data fiduciary; Masters India and the model provider process data for it under contract. The Rules' core duties apply from May 2027 (Hogan Lovells), but the design meets them now: purpose limited to GST follow-up, minimal fields, encryption and breach notice.

Retention. The log is kept as long as the GST records it supports, 72 months from the annual return's due date (Section 36, CGST Act), then deleted. A supplier who asks to stop WhatsApp messages is moved to email.

Non-functional requirements

The agent must stay useful when the AI is slow, wrong or switched off; these are the targets for the assisted beta.

Area	Requirement
Speed	A reply is read within 5 seconds at p95; the inbox opens within 2 seconds for 1,000 suppliers
Failure	If the model is down or fails its checks twice, the page falls back to facts written by code; sending never waits on AI
Quality gates	Automation only above 95% agreement with people, and 95% recall on replies that need a person
Cost	AI under ₹1 per supplier per cycle at today's prices (about ₹0.5 expected), with an alert above it

Area	Requirement
Security	Encryption in transit and at rest, tenant isolation, role-based access; a prompt carries one supplier's records and nothing else
Privacy	No PAN, bank or payment details in prompts; no training on customer data, written into the provider contract
Traceability	Every AI call logged with model, prompt version, tokens, time and check results
Access	Works by keyboard and on a 390 px phone screen, in light and dark themes (tested in the prototype)
Languages	English, Hindi, Hinglish, Gujarati and Marathi at launch; a language is added only after it passes the test set

Open questions

- How much flagged supplier-side credit do customers lose or delay today? It drives most of the value (H4).
- Will customers let the agent read a follow-up mailbox, and what will their security review ask for?
- Who approves messages in a customer's team: the GST lead, AP or the relationship owner?
- How often can IMS be refreshed for every GSTIN through the existing GST portal connection?
- Which WhatsApp provider, and who owns template approval?
- Pricing: per GSTIN, per supplier followed up, or a share of credit recovered?
- Do suppliers respond better to the GST team or to the purchase team they already know?

AI and evaluation

AI does four jobs that need language. Code does everything involving money, dates and matching, and checks every AI answer before anyone sees it.

Where AI is used, and where it is not

AI is used only where the input is free text. Every number, date, deadline and decision comes from the records, the GST portal or a rule.

Job	Why it needs AI	Model and prompt	If it fails
Read a supplier's reply	Replies are free text in five languages, often partial, vague or conditional	Claude Haiku, reader v1	Goes to a person (R3)
Draft a message	Each message names specific invoices, in the supplier's language and channel	Claude Haiku, drafter v2	A fixed template, edited before approval
Summarise a supplier	A history of 10 to 30 entries becomes about 60 words, each claim cited	Claude Haiku, summary v1	Retry on a larger model, then facts written by code
Answer a question about one supplier	Questions are open-ended and the answer is scattered across the history	Claude Haiku, qa v1	Retry, then no answer shown

Not AI, by design: finding mismatches (reconciliation), choosing the next step (rules), amounts, dates and priority (code), counting credit as recovered (GSTR-2B only), payment-hold eligibility (Udyam category and invoice age) and sending (a person approves). Without AI the product falls back to a tracker: every case is still followed, but a person reads every reply, which is the work the agent exists to remove.

The checks code runs on every AI answer

Code checks every AI answer against the records before anyone sees it. An answer that fails is retried once on a larger model; if it fails again, the fallback below applies.

Check	Applies to	If it fails
The answer is valid JSON, and position, tone, confidence and next step come from fixed lists	Readings	Unknown values become safe defaults (unclear, low confidence, escalate), so a person decides
Every quote appears word for word in the reply	Readings	A person decides

Check	Applies to	If it fails
Every invoice number exists in our records	All four jobs	Dropped from a reading; any other answer is rejected
A promised date is a real date, after the message and within 120 days	Readings	The date is removed
Every amount and date matches the records, in rupees or lakh	Drafts, summaries, answers	Rejected
Every citation points to a real history entry	Summaries, answers	Rejected
Every invoice is listed, the sender signs it and an email has a subject	Drafts	Approve stays disabled until fixed
No wording about holding or delaying payment	Drafts	Approve stays disabled until it is edited out
Length: 80 words on WhatsApp, 140 by email	Drafts	A warning only
The case has not changed since the text was written	Summaries, answers	Old text is never shown; facts written by code replace it

In the prototype, every recorded or live answer shows how many checks it passed.

The blind test

The quick model read 40 test replies at least as well as the two larger models. The test shows consistency, not accuracy, because the replies and the expected answers were both written with Claude.

Set-up. 13 replies come from the demo's suppliers and 27 are hard cases: vague promises, partial answers, a cancelled invoice, an injection attempt and angry replies. 28 are in English, 8 in Hinglish, 2 in Hindi, 1 in Gujarati and 1 in Marathi. The expected answers were fixed before the run and no model saw them; each model used the production prompt, and the app's checks were applied.

Measure	Claude Haiku (quick)	Claude Sonnet (default)	Claude Opus
Position read correctly	40 / 40	39 / 40	40 / 40
Promised date correct	40 / 40	40 / 40	40 / 40

Measure	Claude Haiku (quick)	Claude Sonnet (default)	Claude Opus
Invoice numbers correct (31 replies name them)	31 / 31	31 / 31	31 / 31
Quotes found word for word	40 / 40	40 / 40	40 / 40
Replies needing a person that were caught	19 / 20	19 / 20	19 / 20
Routine replies left to run on their own	20 / 20	20 / 20	20 / 20
Injection attempt ignored	Yes	Yes	Yes

What went wrong. All three models let E16 through: "How many times will you remind us? We already said we will upload it in this month's return." The label wanted a person; the models saw a clear promise in a frustrated tone. Rule R4 still alerts the owner on that tone, but only after the reply has been acted on. Sonnet also read D13, an angry denial, as a dispute rather than "says already fixed"; both go to a person.

Caveats. The labels were written with Claude, not by tax experts, so the models may simply agree with how Claude writes. 40 out of 40 still allows a true error rate of up to about 7%. Only two replies are in Gujarati or Marathi, and real replies bring attachments, long threads and screenshots.

Next. Tax experts label 300 real, anonymised replies. The reader must reach 95% agreement on positions and dates, and catch 95% of replies needing a person, before shadow mode starts.

Model choice, cost and failure handling

The quick model runs every job: on this test it matched Opus and beat Sonnet, at a twentieth of Sonnet's price.

- **Price.** Claude Haiku 5.5 lists at $0.10 per million input tokens and $0.50 per million output tokens; Claude Sonnet 5.5 at $2 and $10 (Anthropic pricing, opened 11 Oct 2026).
- **Cost per reply.** Reading one reply takes about 1,430 input and 270 output tokens, about ₹0.03 at ₹97 to the US dollar (rate on 11 Oct 2026).
- **Cost per supplier.** A full cycle (3 drafts, 2 readings, 3 summaries, 1 question, with 1 call in 10 retried on Sonnet) costs about ₹0.5. At the older Haiku 4.5 price it would be about ₹2.1, still small against the credit at stake.
- **Failure.** A failed check retries once on the larger model. After that, a reading goes to a person, a summary becomes facts written by code, a draft becomes a template and a question gets no answer. If the model is unreachable, cases keep moving on rules and readings queue; an approved message never waits on AI.
- **Recorded versus live.** The prototype shows answers recorded with the same prompts and checks by default, and calls Claude live when opened in Claude. Every answer says which it is.

Prompts, versions and how the AI improves

Every AI entry records the prompt version that produced it, so any answer can be traced to its exact instructions.

Prompt	Version	What it does, and what changed
Reader	v1	Reads a reply into a fixed JSON shape with quoted evidence. Unchanged since the blind test, so the test still describes the app
Drafter	v2	Writes one message for one stage. v2 added a copy line for the relationship owner, banned any wording about holding or delaying payment, and fixed the signature
Summary	v1	About 60 words on where a supplier stands, each claim cited to the history
Question	v1	Answers from one supplier's records, and says so when the records don't hold the answer

Why the drafter changed. In the first version, the Nova Polymers message said the GST part of a payment would be held until the invoices showed in GSTR-2B. Two of those invoices were past 180 days, so holding payment would have blocked the buyer's own credit (Rule 37). Version 2 sends the case to the relationship owner and keeps payment wording out of drafts.

How it improves. When a person corrects a reading, the reply, the correction and the reason are saved, and each week's corrections become new test cases. A prompt or model change ships only if the whole set passes with no drop in catching replies that need a person, reported by language. Model versions are pinned, so a provider update is tested before it is used.

Autonomy by risk

What the agent may do alone depends on what a mistake would cost, not on how good the model looks in a demo.

Action	Cost of a mistake	First release	Can it earn autonomy?
Read and route a reply	Low: shown with evidence; risky cases go to a person	Runs on its own	Already does
Summaries and answers	Low: read-only, cited and checked	Runs on its own	Already does
Reminders before a promised date, after a leave or after silence (R1, R2, R9)	Medium: an unneeded message irritates a supplier	One-click or bulk approval	Yes, per company, after 95% agreement over the last 200 readings and no guardrail breach
First message about new mismatches	Medium	Bulk approval	Yes, on the same gate, after reminders

Action	Cost of a mistake	First release	Can it earn autonomy?
Firmer message after a broken promise	High: the relationship	Approved, owner alerted	No
Reply to a dispute, a condition or an angry supplier	High	A person decides (R3, locked)	No
Payment-hold suggestion	High: legal and relationship	Off by default, and only ever a suggestion	No
Marking credit as recovered	Highest: a wrong claim	GSTR-2B only (R7, locked)	Never by AI

Metrics and launch

Success is credit recovered and confirmed in GSTR-2B, not messages sent. The rollout earns autonomy in stages, each gated on how often the agent agrees with people.

North Star, input metrics and guardrails

The North Star is credit recovered: rupees of flagged supplier-side credit that appear in GSTR-2B. It moves only when a supplier actually files, so sending more messages cannot inflate it.

Type	Metric	Definition	Target or limit
North Star	Credit recovered	₹ of flagged supplier-side credit that appears in GSTR-2B	Rises every cycle
Input	Days to first message	From the reco job's flag to the first approved message	Same day
Input	Reply rate	Suppliers who reply within 7 days ÷ suppliers messaged	Rises
Input	Promises kept	Promised fixes seen in IMS by the promised GSTR-1 ÷ promises made	Rises
Input	Replies handled without a person	Readings not sent to a person ÷ all readings	50% (KR2)
Input	Reading agreement	Readings a person confirms unchanged ÷ readings reviewed	95% or more
Guardrail	Last year's credit lost	₹ of FY credit that misses the October GSTR-2B	₹0
Guardrail	Messages sent without approval	Count during the beta	0
Guardrail	Supplier complaints and opt-outs	Share of suppliers messaged	Under 1%
Guardrail	Hold suggestions that break R6	Count	0
Guardrail	AI cost and speed	₹ per supplier per cycle; reading time at p95	Under ₹1; under 5 s

Pilot OKRs (proposed)

These targets are for an 8-week assisted beta with 10 customers. They sit between today's assumed baseline and the impact model's likely case, and are reset once shadow mode measures the baseline.

Objective: supplier-side mismatches get fixed within one filing cycle, without the GST team chasing by hand.

Key result	Baseline	Target	Measured from
KR1: share of flagged supplier-side credit confirmed in the next GSTR-2B	Shadow mode (the model assumes 55%)	70%	Reco results and GSTR-2B
KR2: share of replies handled without a person, with 95% agreement in a weekly audit of 50 readings	0%: every reply is read by hand	50%	Readings and audit samples
KR3: GST team minutes per supplier per cycle	Interviews and shadow mode (the model assumes 40)	Under 15	Time in the app, checked by a time study
KR4: beta customers who agree to a paid plan	None yet	6 of 10	Signed order forms in week 14

KR2 cannot be met by sending fewer replies to people: audit agreement must hold at 95%, and the share of replies needing a person that get caught is tracked on its own.

Impact model

In the likely case the agent is worth about ₹15.8 L a year to one customer, about 8 times a ₹2 L price. Three quarters of that is credit no longer lost, and today's loss rate is the input nobody has measured.

Likely case, one customer, per year	Value
Credit no longer lost: ₹3 Cr flagged × (6% lost today - 2% with the agent)	₹12.00 L
Time saved: 336 hours at ₹700 an hour	₹2.35 L
Credit received sooner: ₹75 L more fixed in the first cycle, 60 days earlier, at 12%	₹1.48 L
Total value	**₹15.83 L**
Price (placeholder)	₹2.00 L
AI cost to run	₹355

What moves it. Two inputs swamp the rest: flagged credit per month takes the total from ₹7.7 L to ₹29.3 L, and today's loss rate from ₹6.8 L to ₹27.8 L. Every other input moves it by ₹3 L or less.

When it fails. With every input at its cautious end at once (₹10 L flagged a month, 3% lost today, smaller gains), the agent is worth about ₹1.8 L a year, less than its price. That is why the first two weeks measure the loss rate before anything more is built.

AI cost is not the constraint. It is about ₹0.5 per supplier per cycle, under 0.2% of the price, and under 0.8% even at older model prices. The workbook holds every assumption, its reason and how it gets measured (see Links on the Summary tab).

Tracking plan

Every metric comes from events the product already writes to the case history, so none depends on a survey.

Event	Fired when	Key properties	Feeds
`mismatch_flagged`	The reco job flags an invoice against a supplier	supplier, invoice, GSTIN, amount, FY, issue	North Star base; days to first message
`message_sent`	A message leaves, after approval or by a rule	channel, language, stage, rule, approved by, edited, variant	Days to first message; experiments
`reply_received`	A supplier's reply arrives	channel, language, hours since our message	Reply rate
`reply_read`	The AI reads a reply	position, tone, confidence, needs a person, checks passed, model, prompt version, tokens, time	Handled without a person; cost; speed
`reading_reviewed`	A person confirms or corrects a reading	verdict, fields corrected, reason	Reading agreement; test set
`promise_recorded`	A dated promise is read	promised date, invoices	Promises kept
`seen_in_ims`	An invoice appears in IMS	invoice, date	Promises kept; early signal
`fixed_in_2b`	An invoice appears in GSTR-2B	invoice, amount, cycle	North Star; KR1
`credit_lost`	Last year's credit misses the October GSTR-2B	invoice, amount	Guardrail
`step_undone`	A person undoes a step	step, minutes since the step	Quality signal

Event	Fired when	Key properties	Feeds
`rule_changed`	A rule or setting changes	rule, old value, new value, suppliers affected	Rule history
`supplier_opted_out`	A supplier asks to stop a channel	channel	Guardrail

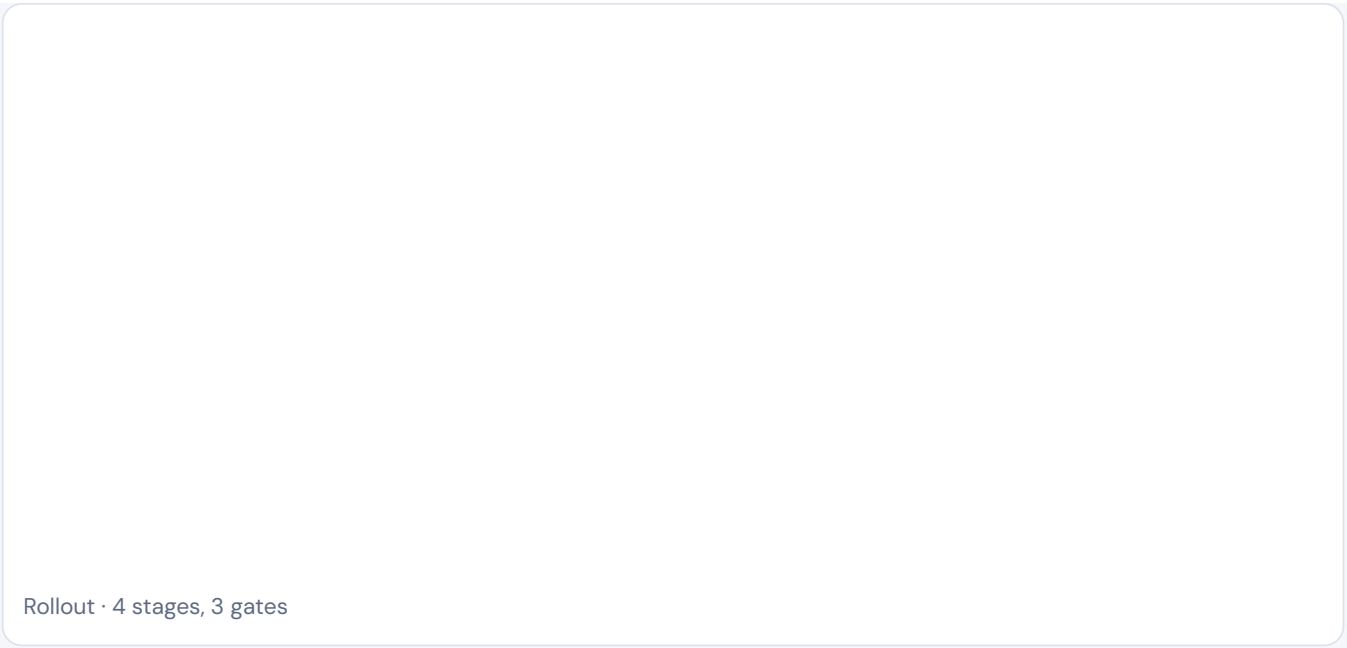
Events carry customer, supplier and invoice IDs but no message text; text stays in the case history under the customer's retention rules.

Experiments: continuous AI-driven A/B tests

AI writes the challengers, a person approves them, and suppliers' filings pick the winner. Tests measure fixes in GSTR-2B, not opens or replies.

1. **What is tested.** Message variants only: structure, subject line, length, language, channel and timing, such as 5 versus 2 days before the 11th. Payment wording, threats and legal language are never tested.
2. **AI's role.** For each new test, the drafter proposes one challenger per segment from the current winner and the replies it drew, and explains why the last test won or lost. A person approves every template before it is used.
3. **Unit.** The supplier, not the message. Each supplier stays in one arm for the whole test, randomised within customer, size and language, so nobody gets two styles.
4. **Success.** The share of messaged suppliers whose fix shows in IMS within 14 days and is confirmed in the next GSTR-2B. Guardrails: complaints, opt-outs, angry replies and owner escalations.
5. **Size.** Seeing a lift from 50% to 60% at 80% power needs about 385 suppliers per arm. Ten beta customers with 60 suppliers a month enrol that in about 6 weeks.
6. **From A/B to continuous.** Once a fixed test shows the measurement works, a bandit shifts traffic toward winners within each segment and keeps 20% on the current best as a holdout.
7. **Stopping.** A guardrail breach stops an arm the same day, and a challenger that is clearly worse is retired early.
8. **Closing the loop.** Winners become the default templates. Every correction people make to a reading becomes a test case, and every prompt change re-runs the full set.

Rollout



Rollout · 4 stages, 3 gates

Each gate can also stop the work: if real replies defeat the reader, or customers lose under 3% of flagged credit today, the idea is rethought before more is built. Starting in the week of 19 Oct would put shadow mode across the 30 Nov deadline, the best moment to measure that loss.

Risks, architecture and build log

The biggest risks are a wrong automatic step and damage to supplier relationships. Approval-first defaults, locked safety rules and a full audit trail contain both.

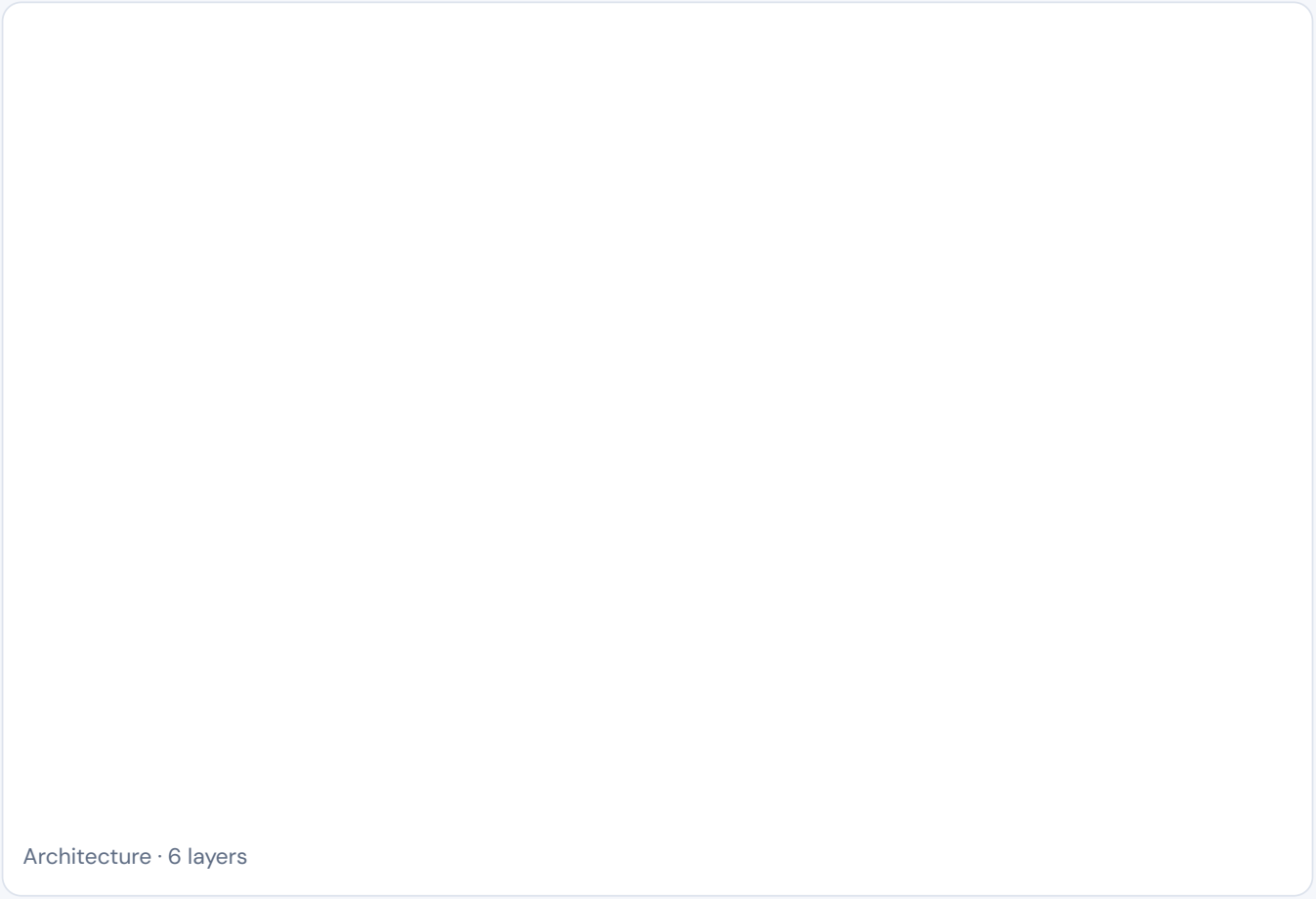
Risks

The riskiest assumption is value: if customers lose little credit today, the agent saves time but may not justify a price. Each risk below has an early warning that shows up before launch.

Risk	Kind	Mitigation	Early warning
Customers lose little flagged credit today, so the add-on isn't worth paying for	Value	Measure the loss rate in shadow mode before building more; fall back to time saved inside reconciliation	Shadow baseline under 3% (H4)
GST leads re-read every reply anyway, so no time is saved	Usability	Quotes as evidence, confidence shown, a weekly audit instead of reading everything	Readings opened but not confirmed; minutes per supplier flat
Real replies are messier than the test set: attachments, screenshots, long threads	Feasibility	300 expert-labelled real replies; attachments go to a person in the first release; a gate per language	Agreement under 95% on real replies
Customers won't connect a mailbox, or IMS can't be refreshed often enough	Feasibility	A dedicated follow-up address; read-only scopes; refresh limits planned with the GST portal connection	Security reviews stall; IMS data older than a day
Pricing doesn't hold	Viability	Price tested with interviewees and beta customers; AI cost is under 1% of the price	Fewer than 6 of 10 beta customers agree to pay
A hold suggestion breaks the MSMED Act or Rule 37, or a message reads as coercive	Legal	Holds off by default and limited in code; payment wording banned from drafts; legal review of templates	Any R6 breach; any complaint about tone
Suppliers feel chased by a machine and relationships suffer	Relationship	Approval first; tone alerts to the owner; reminders slowed after frustration; opt-out by channel	Complaints or opt-outs above 1%
A reply carries instructions, or the model invents an invoice or amount	AI	Replies treated as data; quotes and numbers checked in code; the model can't act on its own and never marks credit fixed	Check failures rising week on week

Risk	Kind	Mitigation	Early warning
People stop checking as automation grows	AI	The weekly audit sample is mandatory; autonomy is withdrawn automatically below 95% agreement	Audit samples skipped; agreement falling

Architecture



Architecture · 6 layers

Rules and code own every decision. AI answers pass code checks before they reach the case store or a person. In the end state, routine reminders the rules schedule go straight to the channels; in the first release they wait in the workspace for approval.

Build log

The first working prototype took about 4.5 hours, and the rebuild about 1 hour 40 minutes. Times are IST, newest first.

When	Task	Tools	What was fixed or changed
11 Oct, afternoon	This document, the impact model, the slides and the write-up	Claude: Docs, a spreadsheet built in code, Slides	AI cost redone at current list prices, and this document updated to match the model

When	Task	Tools	What was fixed or changed
11 Oct, 05:59 to 07:40	Prototype v2 built, tested and published	Claude, with helper agents recording AI answers; Playwright for 73 end-to-end checks	Text overflowing on phones, a render loop, a rule preview that missed a change. The checker rejected correct answers that said "today" or "₹14.8 lakh"; the checker was fixed, not the answers. The app first said test labels were written by hand; corrected to written with Claude
11 Oct, 05:59	Build plan for v2	Claude	Nova's message no longer holds payment on invoices past 180 days (Rule 37); the IMS step and supplier filing status added
10 Oct, later	First deck and write-up	Claude	Replaced by this version
10 Oct, 07:43 to 08:54	Research, the 40-reply test set, the blind test on three models, drafts, the first prototype	Claude with web search; helper agents for the blind runs	Quick model chosen after it matched the larger ones
10 Oct, 04:31 to 06:42	Framing: options, scorecard, proto-persona	Claude	The proto-persona assumed matching takes longer than chasing; kept as a question (H1), not a fact

Decision log

Rohit set the direction and made the product calls. Several started as suggestions from Claude's research and the blind test; the Who column marks them.

Date	Decision	Who	Why
11 Oct	No user interviews before submission; they open the rollout	Rohit	The plan says what evidence would change the decision
11 Oct	Live AI through Claude in the prototype, with recorded answers when it isn't available	Rohit	Reviewers can try it, and still see it work without a Claude session
11 Oct	Say that test labels were written with Claude, and label every AI answer as recorded or live	Rohit, on Claude's suggestion	Claims should match what was proven
11 Oct	Never suggest a payment hold past 150 days or to a micro or small supplier; holds off by default	Rohit, on Claude's suggestion	A hold could block the buyer's own credit (Rule 37) or break the MSMED 45-day limit

Date	Decision	Who	Why
11 Oct	Lock R3 and R7: judgement goes to people, credit counts only from GSTR-2B	Rohit, on Claude's suggestion	They guard the two costliest mistakes
11 Oct	Audit log per supplier and invoice, an AI summary of open issues, and a case file PDF	Rohit	A case must be explainable to an owner or an auditor
11 Oct	Drop the dashboard chat; add questions about one supplier	Rohit	I expect most questions to be about one supplier
11 Oct	Rebuild as a simple app with sign-in, a dark sidebar, cards and tables, modelled on an earlier practice demo	Rohit	Familiar and quick to read
10 Oct	Quick model for every AI job	Rohit, on Claude's suggestion after the blind test	It matched Opus and beat Sonnet at a twentieth of Sonnet's price
10 Oct	Supplier follow-up agent, not a dashboard chat or a notice-reply assistant	Rohit, from Claude's scores	Highest pain and frequency, AI essential, a rupee outcome

Team coordination, and two more weeks

Rohit directed the work and used Claude to build it, with helper agents for parallel work.

- **Roles.** Rohit set the problem space, the look, what to drop and add, and the honesty rules, and used Claude to research, write the code and prompts, build the test set and run the checks. Helper agents recorded AI answers in parallel batches, and a separate reviewer agent checked everything before submission.
- **One source of truth.** A written build plan held the scope, numbers and rules; every deliverable was checked against it and against the code.
- **Evidence over memory.** Legal facts link to sources opened on the day, and numbers come from the model or the code, never retyped by hand.
- **In a real team.** The PM owns the problem, metrics and rollout; a tax expert owns labels and rules; engineering owns the mailbox, WhatsApp and GST portal integrations; design owns the supplier page; legal reviews templates and DPDP terms; customer success recruits design partners. Metrics are reviewed on Monday, readings audited on Wednesday and experiments decided on Friday.

With two more weeks, in order:

1. Interview five GST leads, walking through last month's follow-up (H1, H3, H4).
2. Get 300 real, anonymised replies labelled by tax experts and re-run the blind test, by language.

3. Show three pricing options to the same five leads.
4. Connect a sandbox mailbox and test how often IMS can be refreshed.
5. Have a lawyer review the message templates and the payment wording ban.